ICT2142 – E-Business Systems
Mini Project Final Evaluation Report

“E-Commerce Platform for nr accessories.lk”
Smartphone Accessories Sri Lanka

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1. Introduction

1.1 Background of the Study

The last 10 years have seen the retail trade globally revolutionized by electronic commerce, moving consumer transactions from brick-and-mortar stores to omnichannel digital touch points. With the fast penetration of high speed 4G / 5G mobile broadband in Sri Lanka along with the increasing usage of digital wallets and central bank approved payment gateways, there has been significant demand for online consumer electronics. Within consumer electronic segments, smartphone accessories represent one of the fastest-moving, high-turnover categories of consumer goods. These include protective shockproof phone cases, 9H tempered glass screen protectors, GaN fast chargers, braided USB-C cables, TWS wireless earbuds and smartwatch straps.

1.2 Problem Statement

Even with the surging consumer demand for smartphone accessories in Sri Lanka, the local market remains highly fragmented and continues to grapple with retail challenges:

1. Fake Accessories: Many retail storefronts sell third-party chargers and cables that are of sub-standard quality and lack safety certifications. These can damage the device battery and possibly damage the hardware.

2. Bad Online Buying Experiences: Many domestic accessory websites are clunky, non-responsive interfaces that fail on mobile devices (375px viewports), don't have real-time stock synchronization and have confusing, tedious checkout processes.

3. Bad Order Tracking & Unflexible Payment: Often, consumers are left in the dark about their order status, and websites force cash-on-delivery or use error-prone, insecure payment methods.

1.3 Purpose and Project Scope

The primary purpose of this project is to research, design, architect, and deploy a robust, secure, and modern full-stack web application titled nr-accessories.lk. The platform provides a specialized e-commerce ecosystem tailored for smartphone users across Sri Lanka. The scope encompasses:

A responsive storefront built with modern React.js and Tailwind CSS, featuring an intuitive product catalog, multi-criteria filtering, live keyword search, dynamic product detail views, real-time input validations with instant visual feedback, and a persistent shopping cart.

A secure backend powered by Node.js and Express.js adhering to the Model-View-Controller (MVC) architectural pattern.

A scalable, document-oriented database using MongoDB and Mongoose ODM.

Full integration with the PayHere Sandbox payment gateway for secure local LKR card transactions using backend MD5 hash generation.

A comprehensive, role-protected Store Administrator Management Suite enabling real-time catalog CRUD (Create, Read, Update, Delete) operations, inventory tracking, and customer order fulfillment lifecycle management.

1.4 Objectives of the System

• **User Management:** Implement secure user registration, bcrypt password hashing, stateless JWT session authentication, and persistent customer delivery profile management.

• **Product Management:** Provide dynamic product catalogs with instant keyword search, category filtering, live inventory indicators, and promotional discount calculations.

• **Cart & Checkout:** Engineer a frictionless shopping cart backed by browser local storage caching and responsive client-side form validation.

• **Financial Security:** Integrate the PayHere payment gateway with cryptographic server-side MD5 signature calculation to prevent price tampering.

• **Administrative Control:** Equip store operators with an analytical dashboard to manage products, monitor low inventory levels, and update order statuses.

• **Usability Assurance:** Ensure 100% responsiveness on mobile devices down to 375px viewports without horizontal clipping or broken elements.

2. Business Description

2.1 Business Overview: nr-accessories.lk

nr-accessories.lk is an online specialized retail outlet dealing only in premium genuine smart phone accessories and peripherals. The venture, built on a direct-to-consumer e-commerce model, removes middleman mark-ups and overheads of brick and mortar stores, passing on savings to tech-savvy Sri Lankan consumers while guaranteeing authenticity of products.

Identity of a brand Modern, reliable, customer and technology oriented.

Delivery : Islandwide delivery to all 25 administrative districts of Sri Lanka. Delivery within the Western Province (Colombo, Gampaha, Kalutara) within 24-48 hours.

2.2 Product Catalog and Categories

The product inventory of nr-accessories.lk is structured into five core high-demand accessory categories:

1. Cases & Protective Covers: Heavy-duty MagSafe cases, crystal-clear anti-yellowing TPU cases, slim matte silicone bumpers, and leather folio covers for flagship Apple iPhone, Samsung Galaxy, Google Pixel, and Xiaomi devices.

2. Screen & Camera Lens Protectors: 9H hardness tempered glass, edge-to-edge privacy filters, matte anti-glare screen protectors, and sapphire camera ring protectors.

3. Chargers & Power Adapters: Gallium Nitride (GaN) fast wall chargers (20W, 30W, 65W, 100W PD), Qi-certified wireless charging pads, multi-device charging stations, and high-capacity power banks (10,000mAh to 30,000mAh).

4. Cables & Interconnects: MFi-certified USB-C to Lightning cables, 100W/240W braided USB-C to USB-C fast charging cables, durable audio jack adapters, and OTG adapters.

5. Audio & Wearables: True Wireless Stereo (TWS) low-latency earbuds, active noise-cancelling (ANC) headphones, sports neckbands, silicone smartwatch replacement straps, and protective smartwatch bumpers.

2.3 Target Market and Customer Persona

The primary target market consists of:

• **Tech Enthusiasts & Young Professionals (Ages 18–35):** Consumers who own mid-to-high-end smartphones and prioritize premium protective gear, fast charging capabilities, and authentic branded accessories (e.g., Anker, Spigen, Baseus, Ugreen).

• **University Students & Mobile Gamers:** Price-conscious consumers seeking durable, high-performance cables, low-latency gaming earphones, and long-lasting screen guards.

• **Secondary Market:** Busy corporate workers seeking reliable home delivery and hassle-free online card payments for their mobile accessory replacements.

2.4 Revenue Model and Value Proposition

Revenue is primarily generated through retail profit margins on imported and locally sourced authentic accessory inventory. The key value propositions include:

1. Guaranteed Authenticity: 100% genuine products backed by a 6-month replacement warranty on charging equipment and adapters.

2. Free Delivery Threshold: Complimentary islandwide courier delivery on orders exceeding Rs. 7,500.

3. Payment Flexibility: Support for online credit/debit card payments via PayHere Sandbox, Buy-Now-Pay-Later (Koko roadmap), and Cash on Delivery.

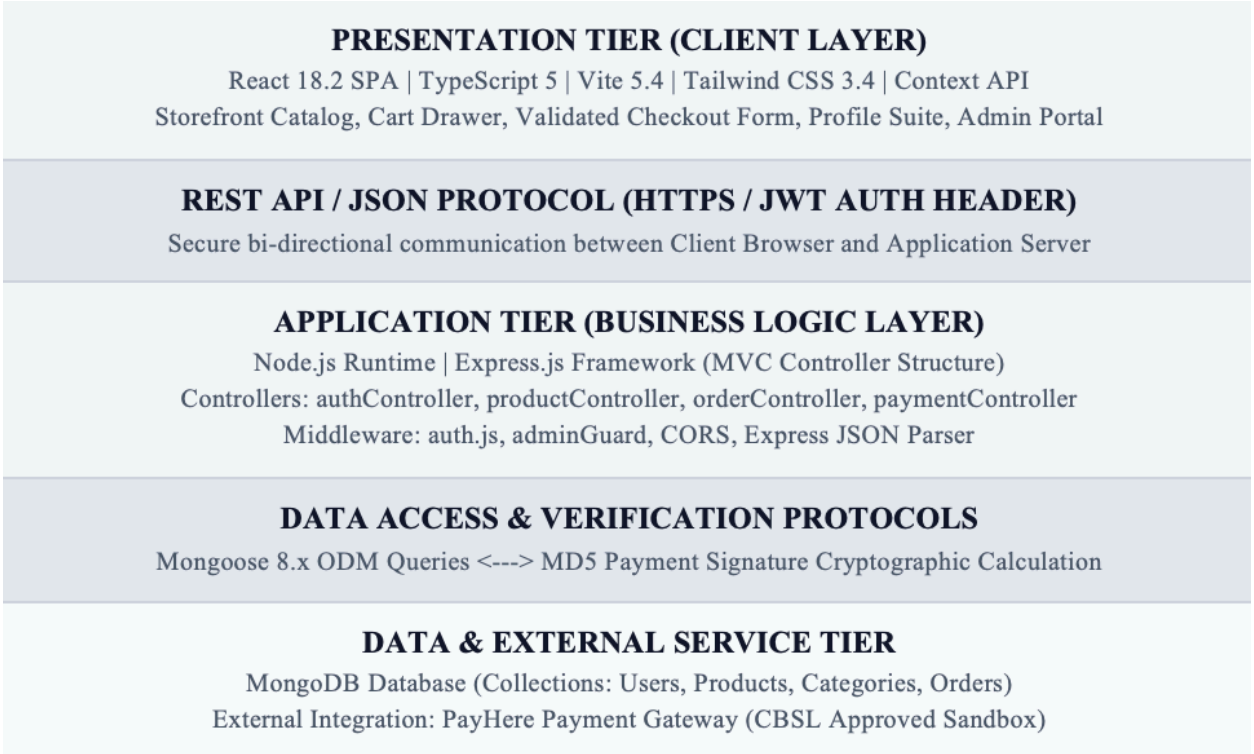
2.5 Competitive Advantage in Sri Lankan Market

Unlike generic e-commerce marketplaces (e.g., Daraz) where finding high-quality smartphone accessories can be tedious due to counterfeit duplicates, nr-accessories.lk provides a curated, vetted inventory with clear compatibility tables, a lightning-fast ad-free interface built on modern Single Page Application architecture, and real-time automated stock tracking ensuring customers never order out-of-stock items.

3. System Architecture

3.1 Architectural Overview (3-Tier MVC Pattern)

The nr-accessories.lk platform utilizes an industry-standard Three-Tier Architecture, separating concerns across Presentation, Business Logic, and Data Persistence layers. This decoupled structure guarantees modularity, maintainability, and horizontal



3.2 Frontend Client Layer

The frontend is constructed as a high-performance Single Page Application (SPA) utilizing React 18.2, TypeScript 5.0, and Vite 5.4. The layout is designed mobile-first using Tailwind CSS 3.4. Global application state—such as user authentication sessions, role permissions, and shopping cart persistence—is orchestrated using the React Context API (AuthContext and CartContext) coupled with browser localStorage caching.

3.3 Backend Application Layer

The backend is powered by Node.js running an Express.js RESTful API server. The API strictly follows the Model-View-Controller (MVC) paradigm:

• **Routes:** Define URL route endpoints (/api/products, /api/orders, /api/auth, /api/payment) and route incoming requests.

• **Controllers:** Encapsulate business logic, such as catalog filtering, inventory deduction, payment signature generation, and password verification.

• **Middleware:** Handle cross-cutting concerns including JSON parsing, CORS headers, JWT verification (auth.js), and Administrator privilege guards (adminGuard).

3.4 Database and Storage Layer

A MongoDB NoSQL document database acts as the primary datastore, interfaced via Mongoose 8.x Object Data Modeling (ODM). MongoDB was selected for its high write throughput, dynamic schema flexibility for diverse product attributes, and seamless JSON serialization with Express.

3.5 External Integrations (Payment Gateway)

The platform integrates the PayHere Sandbox Payment Gateway (approved by the Central Bank of Sri Lanka). Financial transactions are authenticated via MD5 cryptographic hashing calculated on the backend to guarantee zero client-side price tampering.

4. Tools and Technologies Used for Development

The development of nr-accessories.lk leveraged modern, production-grade open-source tools across all architectural tiers. The table below summarizes the chosen technology stack along with technical justifications:

Layer / Category	Technology	Version	Technical Justification & Purpose
Frontend Framework	React.js	18.2.0	Declarative component-based UI delivering fast Virtual DOM reconciliation and responsive state binding.

Language	TypeScript	5.0+	Strongly-typed superset of JavaScript preventing compile-time bugs and enforcing data schemas.
Build Tool	Vite	5.4	Ultra-fast Hot Module Replacement (HMR) and optimized ES-module build bundling.
Styling & CSS	Tailwind CSS	3.4	Utility-first responsive styling framework enabling fluid mobile-first layouts without custom CSS bloat.
Iconography	Lucide React	0.344	Crisp, lightweight SVG icons for navigation, cart badges, star ratings, and admin controls.
State Management	React Context API	Native	Centralized global state management for User Authentication and Shopping Cart persistence.
Backend Runtime	Node.js	v18+	Event-driven, non-blocking asynchronous JavaScript execution environment.
Web Framework	Express.js	4.19	Robust REST routing framework supporting middleware chains and MVC controllers.
Database	MongoDB	7.0+	High-throughput NoSQL document database ideal for flexible JSON-like product catalogs.
Database ODM	Mongoose	8.3	Schema validation library enforcing referential integrity and population across documents.
Authentication	JSON Web Tokens	9.0	Stateless cryptographic bearer token authentication for client-server sessions.
Password Security	Bcrypt.js	2.4	Industry-standard salted password hashing algorithm preventing dictionary attacks.
Payment Gateway	PayHere Sandbox	v1.0	Central Bank of Sri Lanka compliant payment aggregator for local card transactions.
QA & Verification	Puppeteer	22.0+	Headless Chrome browser automation framework for automated mobile audits and test validation.

Version Control	Git & GitHub	Latest	Distributed source control for codebase versioning, tracking, and collaboration.

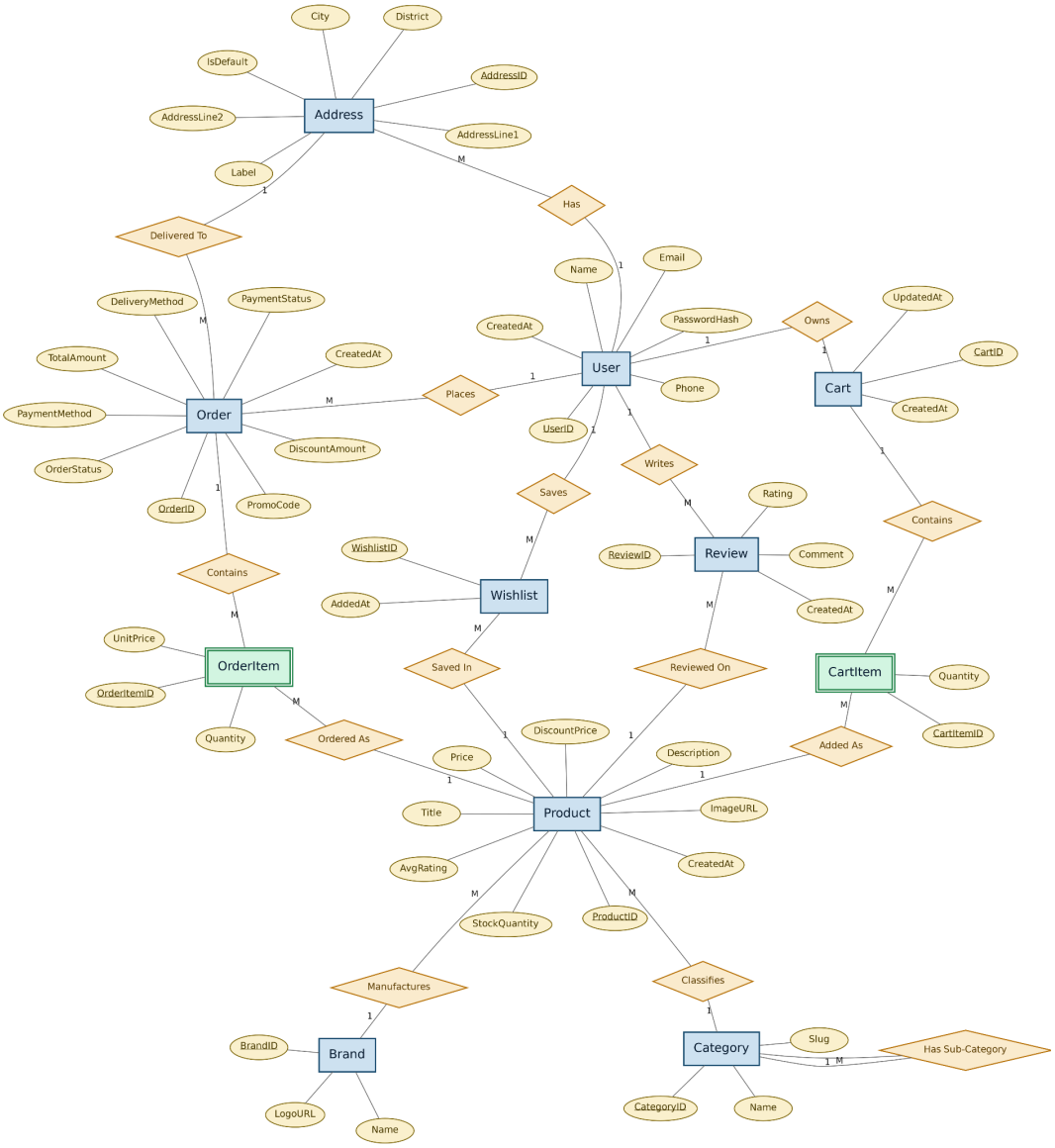
5. Database Design

5.1 Database Architecture Rationale

A document-based database model using MongoDB was chosen because e-commerce catalogs contain semi-structured product specifications (e.g., varying cable lengths, charger wattages, case colors, and warranty terms) that fit naturally into flexible JSON documents. Foreign key references (ObjectId) are utilized to maintain relational links between users, products, categories, and orders.

5.2 Entity-Relationship (ER) Overview

The data architecture comprises four interconnected collections: Users, Categories, Products, and Orders. The entity-relationship model establishes one-to-many associations between Categories and Products, as well as between Users and Orders.



5.3 Detailed Collection Schemas

5.3.1 Users Collection (User.js):

Holds registered customer profiles and store administrator credentials.

• _id: ObjectId (Primary Key, unique document identifier)
• name: String (Required, customer's full name)
• email: String (Required, unique indexed, lowercased)
• password: String (Required, Bcrypt 10-round salted hash)
• role: String (Enum: 'customer' | 'admin', default: 'customer')
• phone: String (Customer contact phone number)
• addresses: Array of Subdocuments [addressLine1, city, district, postalCode]
• createdAt, updatedAt: Date (Automated timestamps)

5.3.2 Products Collection (Product.js):

Stores smartphone accessories, live stock counts, and pricing.

• _id: ObjectId (Primary Key)
• title: String (Product title, e.g., 'Anker 30W GaN Fast Charger')
• brand: String (Manufacturer, e.g., 'Anker', 'Spigen', 'Baseus', 'Apple')
• category: ObjectId (Foreign key reference to Category._id)
• description: String (Comprehensive specifications and compatibility notes)
• price: Number (Regular retail price in LKR)
• discountPrice: Number (Promotional selling price in LKR, optional)
• stockQuantity: Number (Real-time available inventory counter)
• imageURL: String (Local image path or web URL, e.g., '/images/anker_30w.jpg')
• isFeatured: Boolean (Flags product for promotional hero sections)

5.3.3 Categories Collection (Category.js):

Classifies accessories into structured catalog taxonomy.

• _id: ObjectId (Primary Key)
• name: String (Category title, e.g., 'Cases & Covers', 'Chargers & Adapters')
• slug: String (URL-friendly string, e.g., 'cases-covers')
• description: String (Taxonomy overview)

5.3.4 Orders Collection (Order.js):

Captures customer checkout records, payment statuses, and fulfillment.

• _id: ObjectId (Primary Key)

• user: ObjectId (Foreign key reference to User._id)

• items: Array of embedded items [{ product: ObjectId, title: String, quantity: Number, unitPrice: Number }]

• deliveryAddress: Embedded object { fullName, phone, streetAddress, city, district }

• paymentMethod: String (Enum: 'payhere' | 'cash_on_delivery')

• paymentStatus: String (Enum: 'pending' | 'paid' | 'failed')

• orderStatus: String (Enum: 'pending' | 'processing' | 'shipped' | 'delivered' | 'cancelled')

• subtotal: Number, deliveryCharge: Number, totalAmount: Number (Calculated on server)

5.4 Data Integrity, Foreign Keys, and Indexing

To maintain relational consistency in a NoSQL environment, Mongoose population middleware dynamically resolves references. A unique index on User.email prevents duplicate registrations. Stock decrement operations ($inc: { stockQuantity: -quantity }) execute atomically upon payment verification to prevent race conditions.

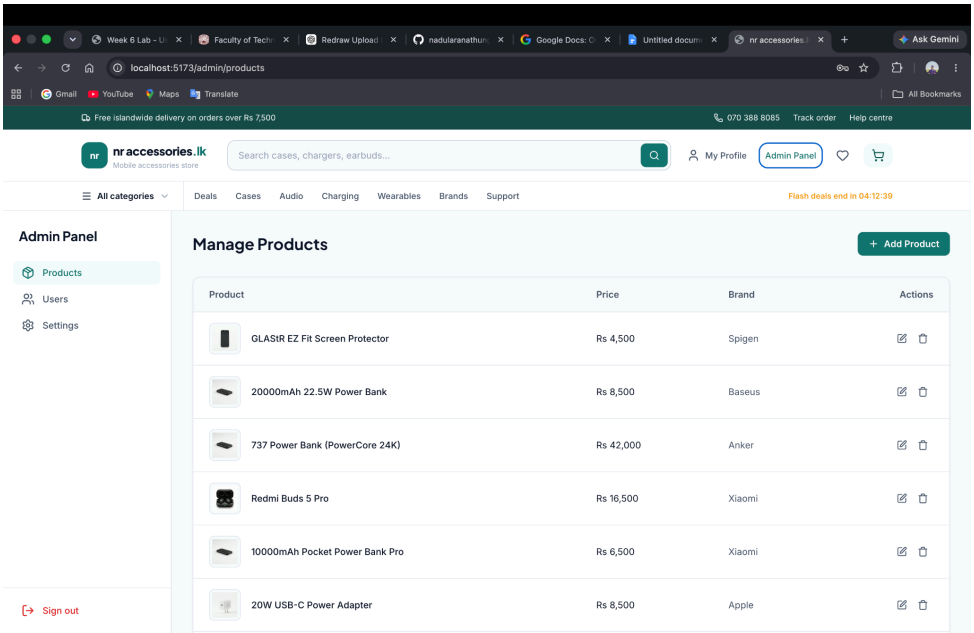
6. User Interface (UI) Screenshots and Functional Walkthrough

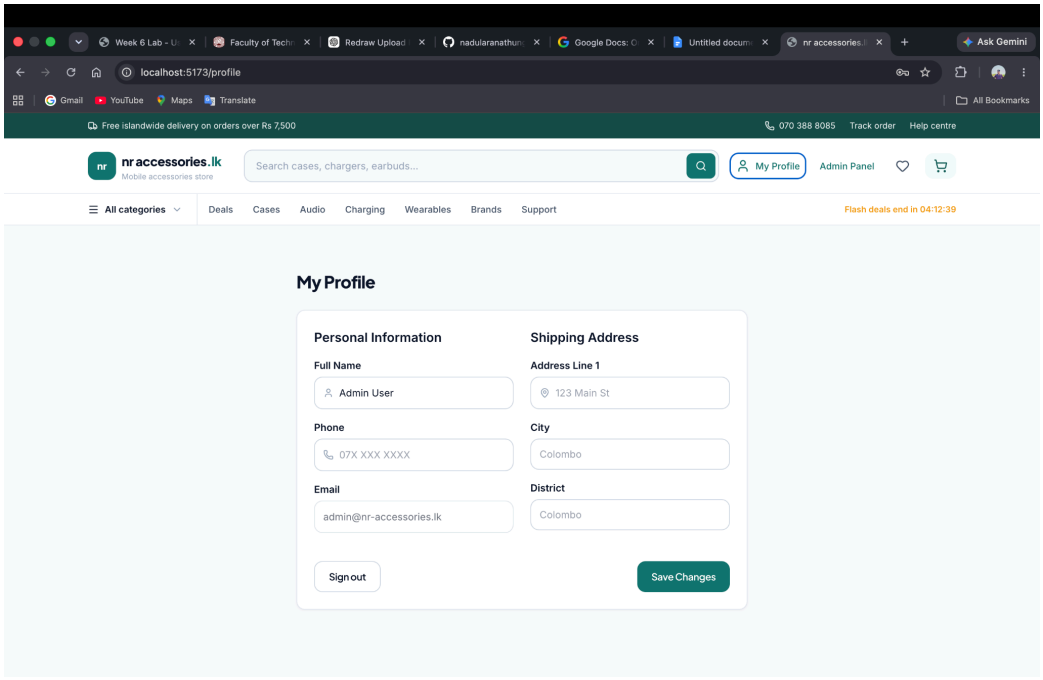
This section demonstrates the complete functional walkthrough of nr-accessories.lk across all mandatory evaluation categories, accompanied by authentic system screenshots.

6.1 User Management Functionalities

The user management module provides a secure onboarding experience. New visitors can register via /register with real-time validation checking email syntax and password length. Existing users authenticate at /login, receiving a signed JWT stored securely in the browser session. Upon authentication, the header dynamically updates to display the customer's name and profile menu.

In the Customer Profile (/profile), users can review their order history and update their personal delivery address (street address, city, district, contact number). Updated profiles are written directly to MongoDB and automatically pre-fill the checkout form on future visits.



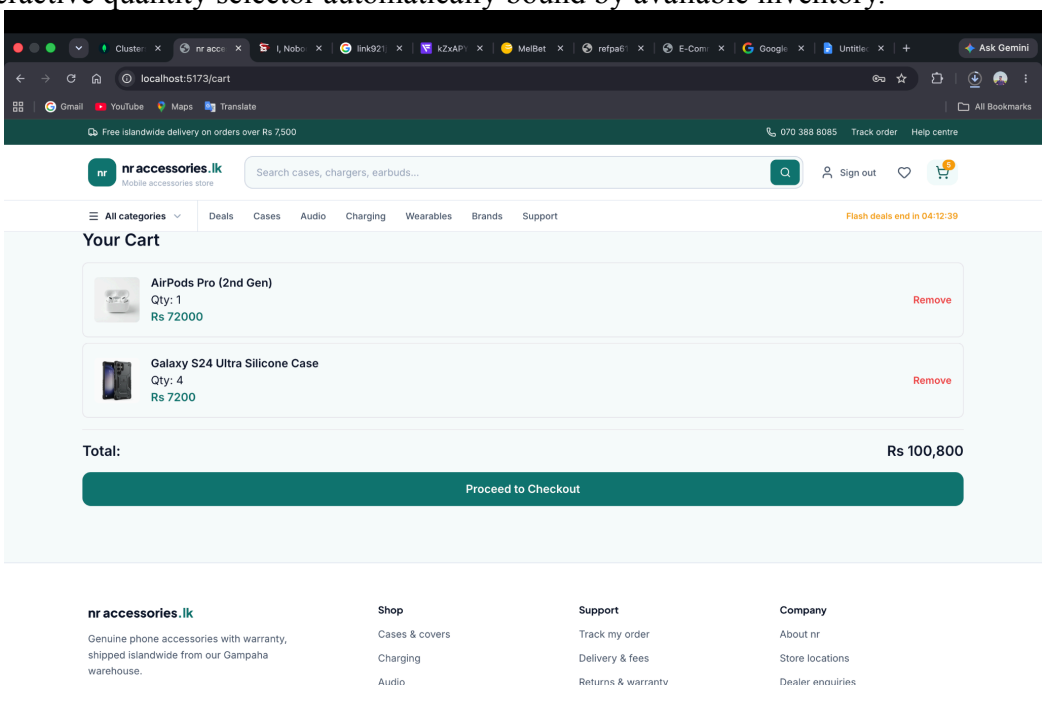


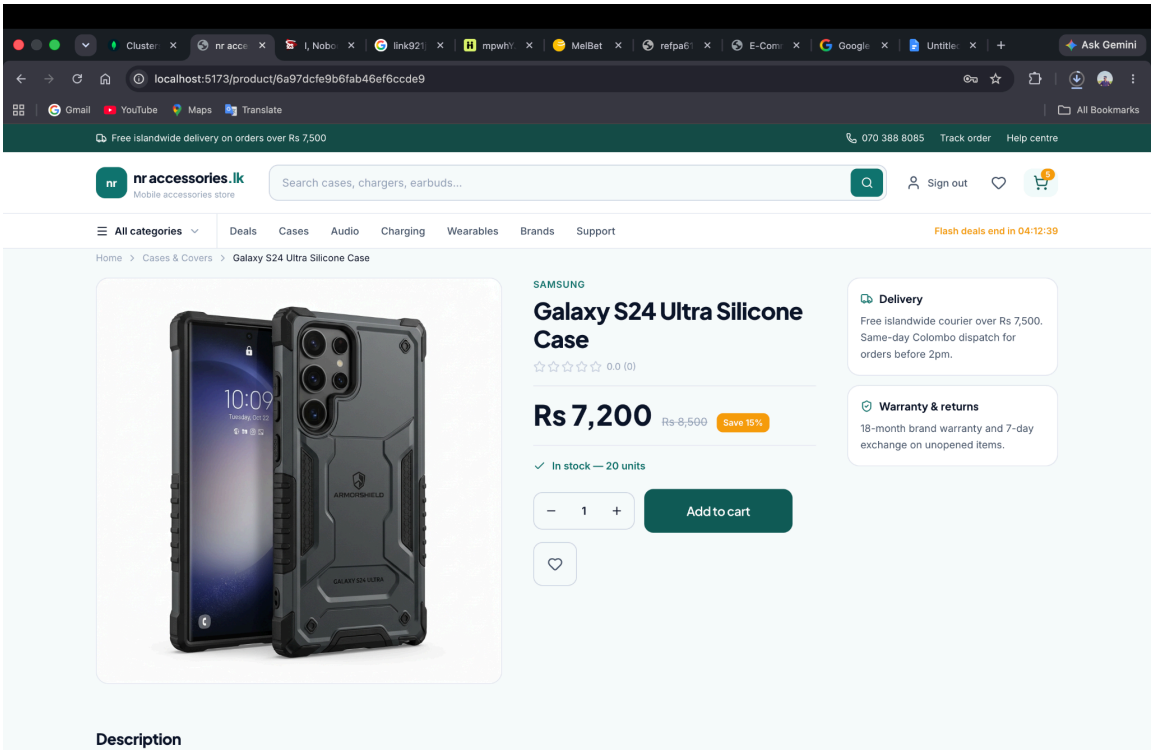
6.2 Product Browsing, Search, and Filtering

The product catalog (/shop) presents all active accessories in a responsive grid layout. Each product card displays a high-resolution image, brand tag, title, regular price, promotional discount percentage, and a live inventory badge (In Stock vs. Out of Stock).

The platform features instant keyword search: typing brand names (e.g., 'Anker', 'Spigen') or product keywords instantly updates the catalog without page reloads. In addition, users can filter by Category (Cases, Screen Protectors, Chargers, Audio), Brand, and Price Range sliders.

Clicking any product opens the Product Details Page (/product/:id), featuring a photo gallery with hover zoom, technical specifications, device compatibility charts, warranty information, and an interactive quantity selector automatically bound by available inventory.





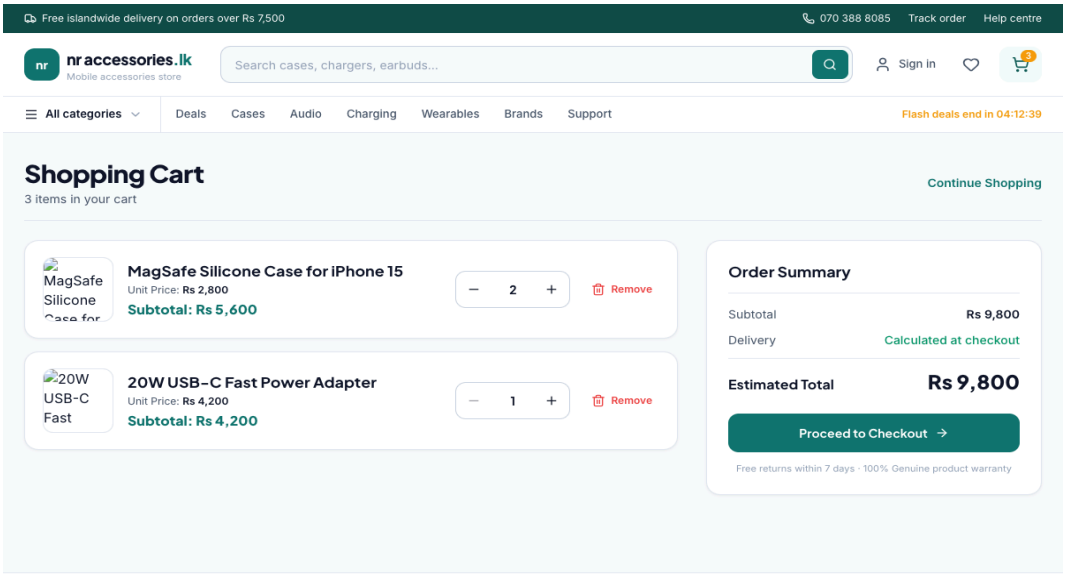
6.3 Shopping Cart System and Local Persistence

Customers can add items to their cart directly from product cards or detail pages. The Shopping Cart is accessible via a slide-over drawer or dedicated /cart page. Key capabilities include:

Dynamic Quantity Steppers: Users can increment or decrement item quantities; line item subtotals, courier delivery fees, and net grand totals recalculate instantaneously.

Item Removal: Items can be removed with a single click, instantly updating total costs.

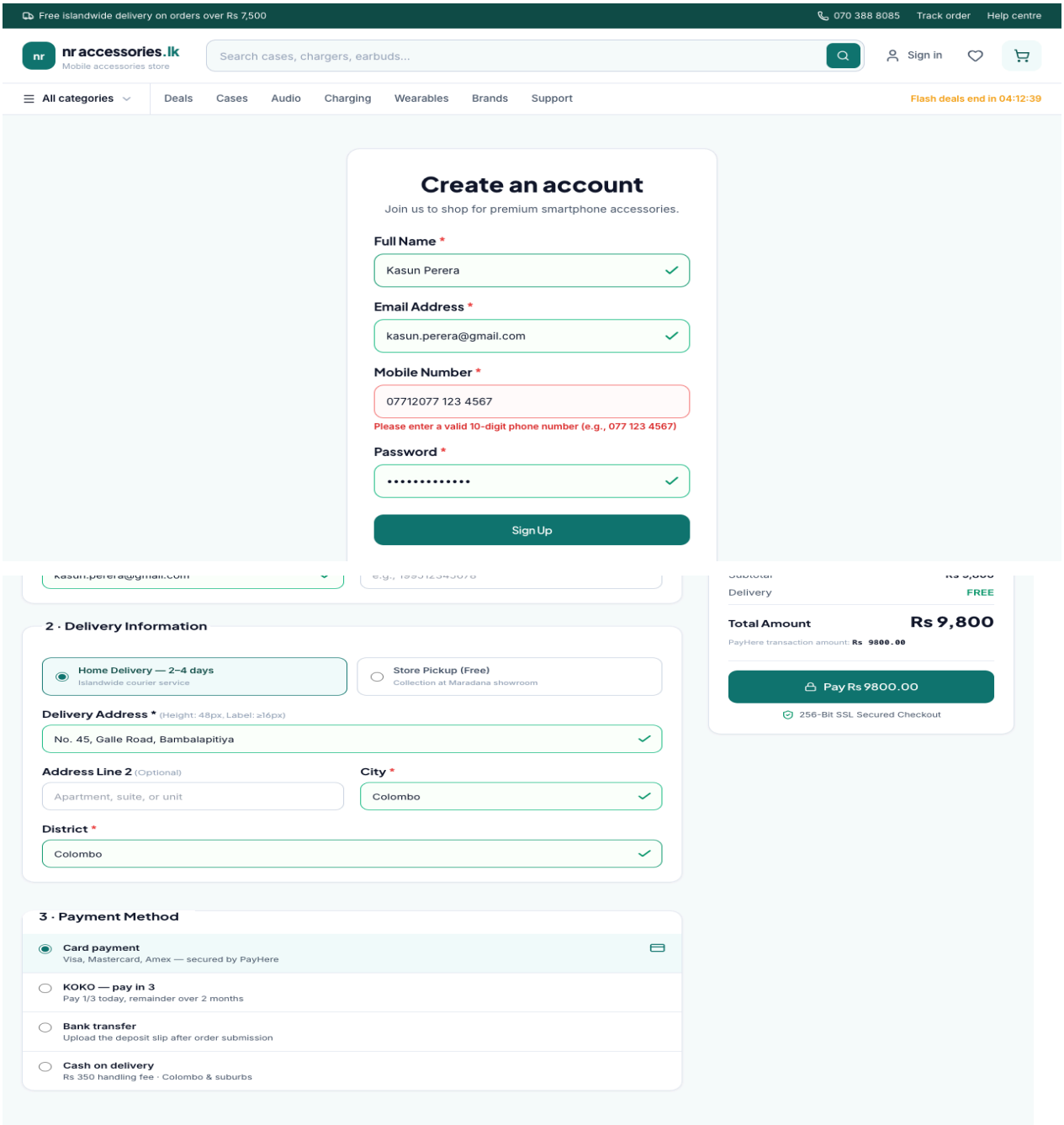
Session Persistence: Cart state is synchronized with localStorage. Even if the customer refreshes the browser or closes the tab, cart items and quantities remain completely intact.



6.4 Checkout Process and Real-Time Form Validation

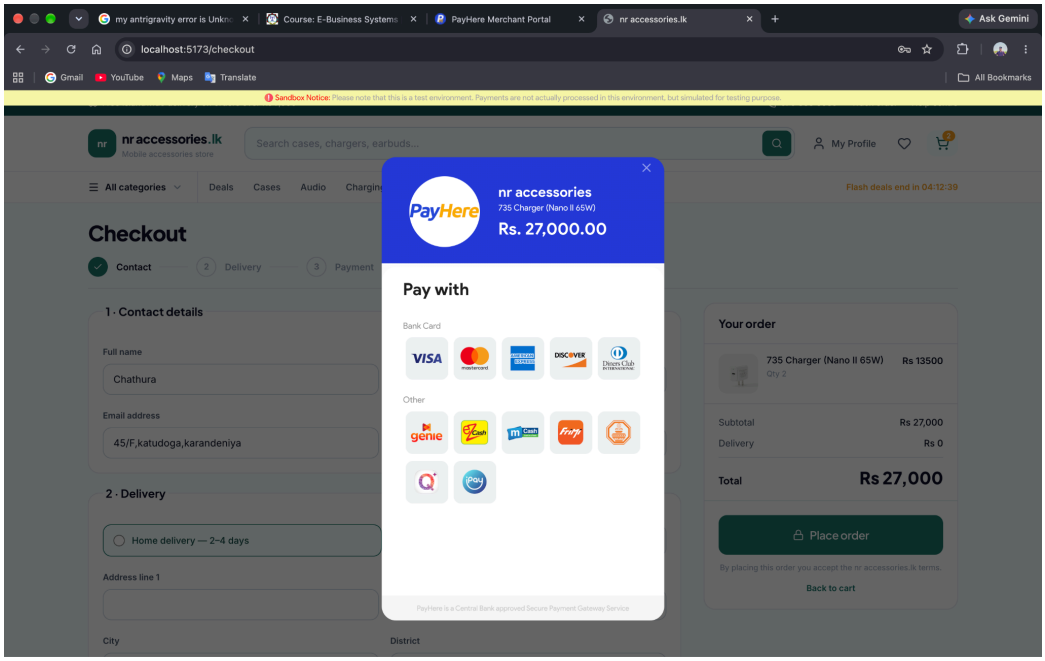
The checkout module (/checkout) adopts a conversion-optimized two-column layout. The right column displays an itemized order summary, including courier charges (free shipping on orders above Rs. 7,500). The left column presents the delivery contact form.

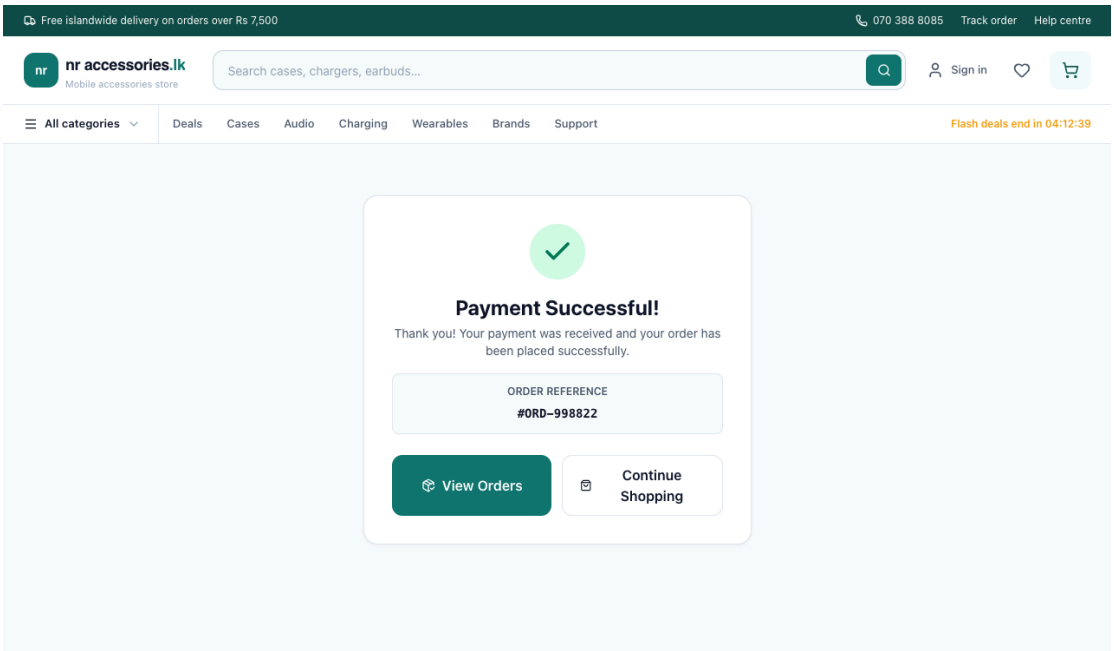
A distinctive usability feature of nr-accessories.lk is Real-Time Visual Validation: as users type their full name, email, Sri Lankan phone number (e.g., 070 388 8085), and street address, fields provide immediate visual feedback with a vibrant green border and a green checkmark (✓). If an invalid entry is made, an alert message gently clarifies the required format prior to submission.



6.5 Order Confirmation and Customer Order History Tracking

Upon successful transaction verification, customers are directed to the Order Confirmation Page (/order-success/:id). The confirmation screen presents the unique Order ID, receipt summary, delivery address breakdown, and estimated delivery timeline. Customers can also revisit their personal Order History under My Profile to track order fulfillment stages.





6.6 Admin Management Suite (Dashboard, Product CRUD, Fulfillment)

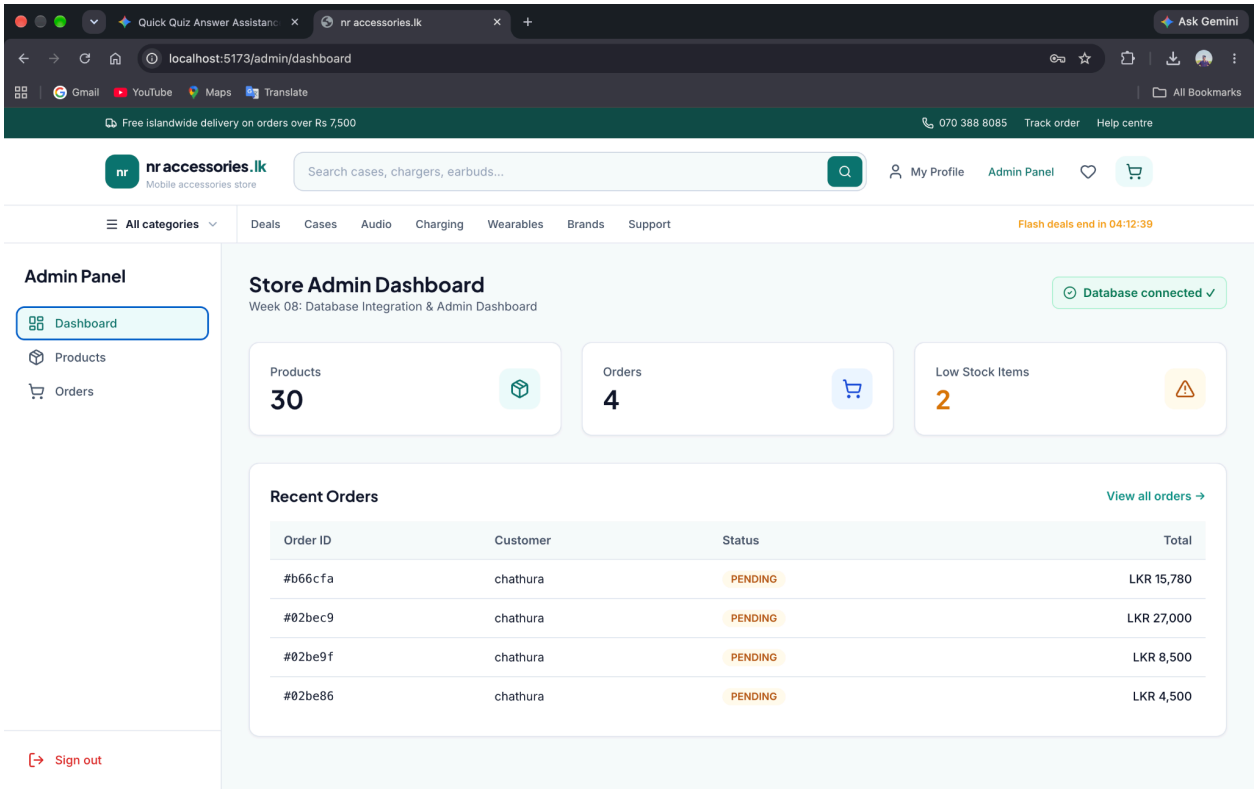
The Store Administrator Suite (/admin) fulfills all administrative evaluation requirements and is protected by role-based JWT route guards:

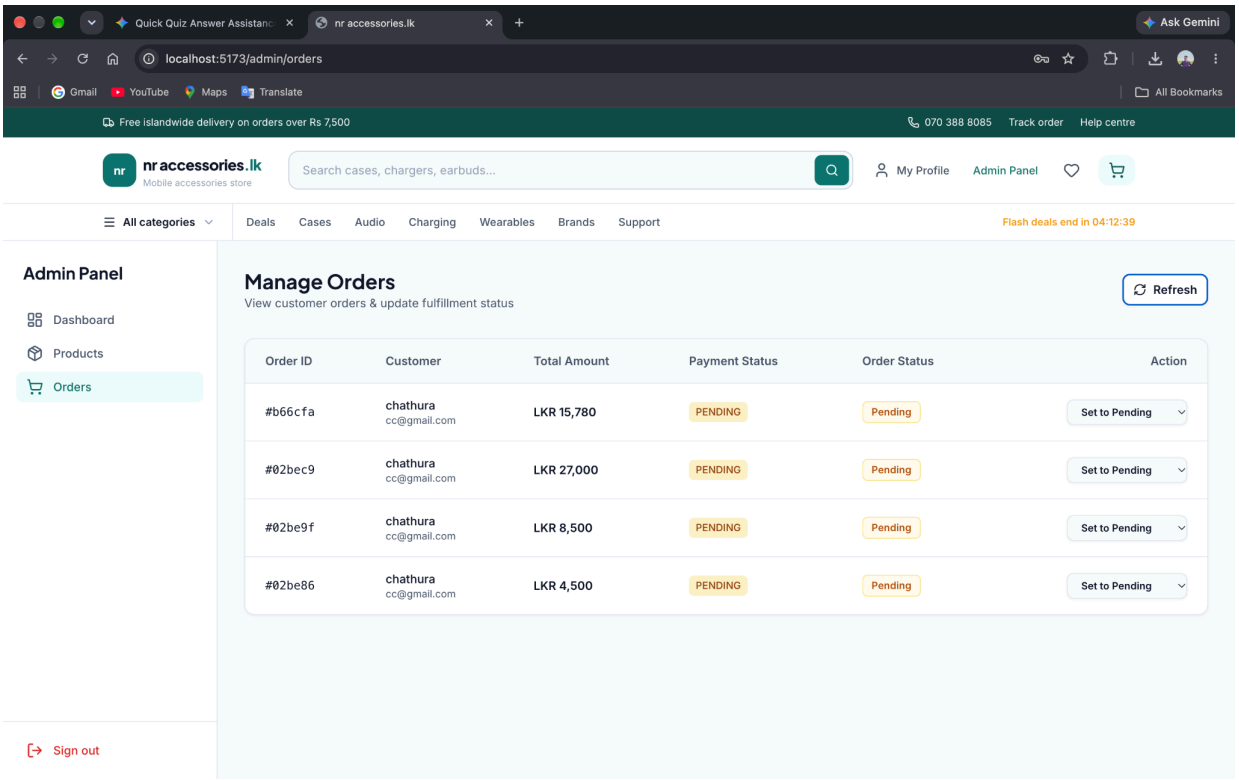
1. Analytical Dashboard Overview: Presents high-level business analytics: Total Revenue, Total Orders Placed, Active Products in Catalog, and a Low Stock Alert counter (stock <= 5 items).

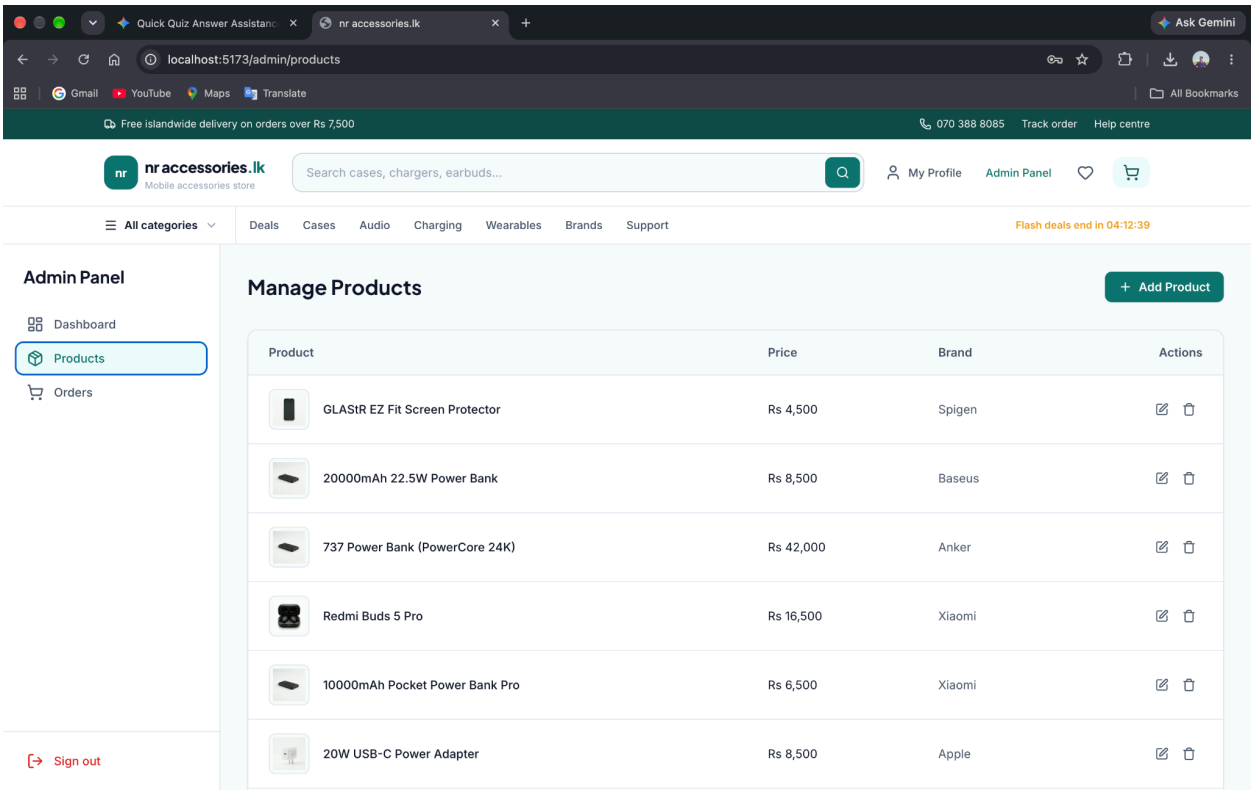
2. Product Inventory CRUD: Admins can create new items using an interactive modal with live image thumbnail previews, edit existing prices and inventory, or safely delete discontinued accessories.

3. Order Fulfillment Management: Admins inspect incoming customer orders, customer contact phone numbers, and delivery addresses, and transition order statuses (Pending -> Processing -> Shipped -> Delivered).

4. User Management: View registered user accounts and manage administrative privileges.

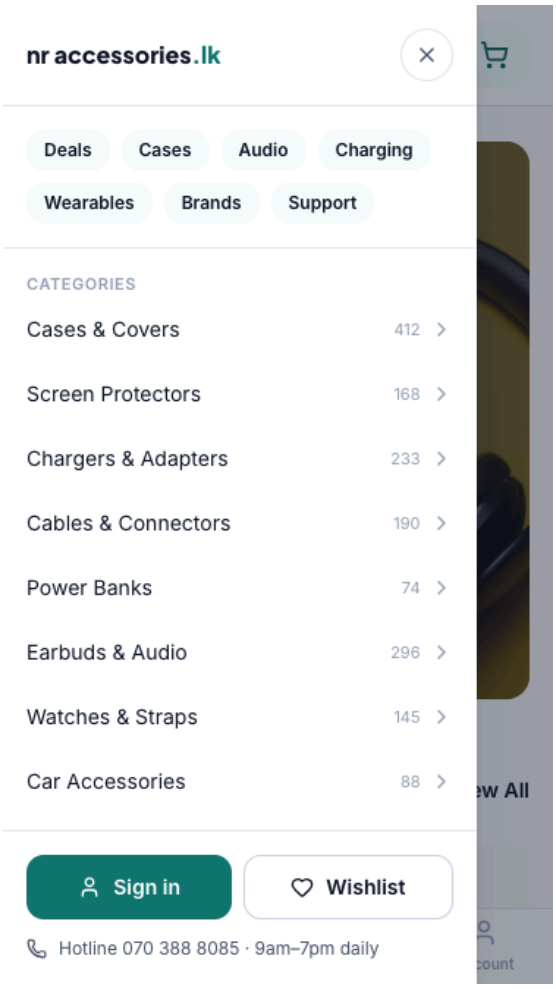


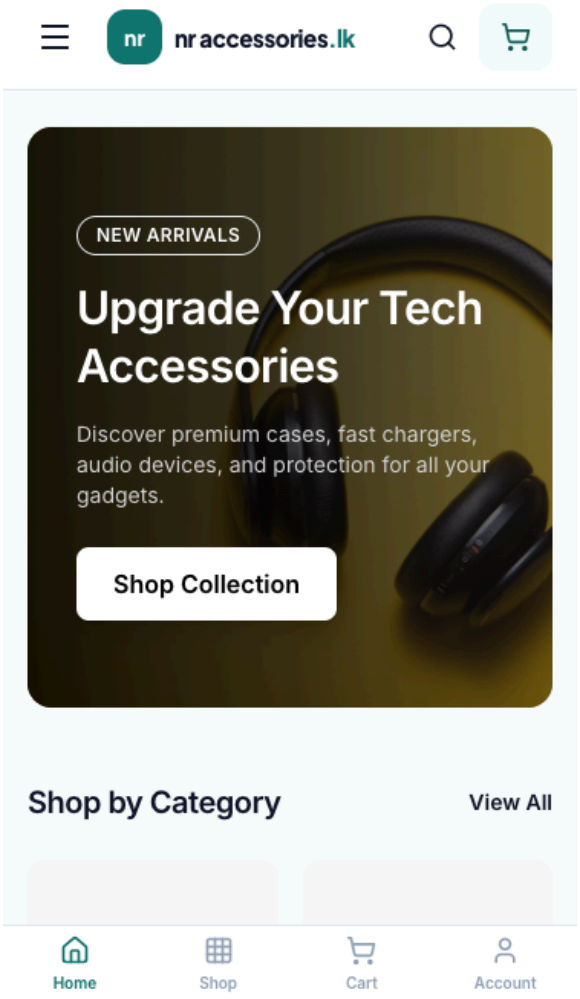

Quick Quiz Answer Assistanc
nr accessories.lk
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Free islandwide delivery on orders over Rs 7,500
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Mobile accessories store
Search cases, chargers, earbuds...
My Profile
Admin Panel
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Dashboard
Products
Orders
Sign out
Manage Products
+ Add Product
Product
Price
Brand
Actions
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Spigen
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Rs 8,500
Baseus
737 Power Bank (PowerCore 24K)
Rs 42,000
Anker
Redmi Buds 5 Pro
Rs 16,500
Xiaomi
10000mAh Pocket Power Bank Pro
Rs 6,500
Xiaomi
20W USB-C Power Adapter
Rs 8,500
Apple

6.7 Mobile Usability and Responsive Viewports (375px+)

Recognizing that the vast majority of e-commerce traffic in Sri Lanka originates from mobile devices, the platform underwent rigorous mobile usability auditing on a 375px viewport (representing Apple iPhone SE, iPhone 13 mini, and standard Android screens). The audit confirmed zero horizontal overflow, fluid touch drawers, and a persistent Bottom Mobile Tab Bar providing single-thumb access to Home, Shop, Cart, and Profile.





7. Security Features Implemented

The nr-accessories.lk platform integrates multiple layers of security to protect consumer personal data and guarantee the integrity of financial transactions:

7.1 Cryptographic Password Hashing (Bcrypt)

Plaintext passwords are never stored in the database. During user registration, the system invokes bcrypt.genSalt(10) to generate a 10-round salted one-way cryptographic hash. Even in the event of an unauthorized database dump, passwords cannot be reversed or cracked using rainbow table attacks.

7.2 Stateless Authentication via JSON Web Tokens (JWT)

User sessions are managed statelessly using cryptographically signed JSON Web Tokens (JWT). When a user logs in, the Express server signs a token payload containing their user ID and role using a secret environment key (JWT_SECRET) with a 7-day expiration. The client includes this token in the HTTP Authorization: Bearer header for protected API calls.

7.3 Role-Based Access Control (RBAC) Route Guards

To prevent unauthorized manipulation of catalog data or administrative views, the backend implements two-tiered middleware guards: auth.js verifies the authenticity of the JWT token, while adminGuard inspects the decoded user role. Non-admin requests attempting to perform CRUD actions on /api/products or view all orders are rejected with HTTP 403 Forbidden.

7.4 Input Sanitization and Dual-Layer Validation

Security validation is enforced across both client and server boundaries. The frontend verifies email syntax and Sri Lankan mobile numbers via regex patterns. On the backend, Mongoose schema validators enforce data types, required constraints, and positive number boundaries (preventing negative prices or negative stock quantities).

7.5 Payment Hash Signature Verification (MD5 Hashing)

To prevent malicious actors from altering order amounts using browser developer tools, payment hashes are calculated strictly on the backend. The server generates an MD5 hash combining the Merchant ID, Order ID, Formatted Amount, Currency, and the hashed Merchant Secret. The PayHere gateway validates this cryptographic signature before accepting payment.

7.6 CORS and Environment Isolation

Cross-Origin Resource Sharing (CORS) middleware restricts API access exclusively to authorized frontend origins. Sensitive credentials—including MongoDB Atlas connection strings, JWT secret keys, and PayHere merchant secrets—are isolated inside server/.env files and excluded from version control via .gitignore.

8. Payment Workflow Explanation

8.1 PayHere Payment Gateway Integration Architecture

PayHere is the leading Central Bank of Sri Lanka (CBSL) approved payment aggregator, supporting Visa, MasterCard, AMEX, and digital banking methods in Sri Lankan Rupees (LKR). The nr-accessories.lk platform integrates PayHere Sandbox via an asynchronous modal popup workflow.

8.2 End-to-End Payment Sequence

The payment transaction proceeds through seven distinct phases:

1. Checkout Initiation: Customer clicks 'Pay with PayHere' on the checkout page.

2. Order Staging: The client sends a POST request with line items and delivery address to /api/orders.

3. Cryptographic Signature: The server calculates net totals, creates a MongoDB order record with status 'pending', and generates an MD5 verification hash token.

4. Modal Launch: The frontend initializes the official PayHere Javascript SDK (payhere.startPayment) with merchant credentials and generated hash.

5. Card Verification: The customer enters sandbox card details (4111 1111 1111 1111), expiry date, CVV, and OTP.

6. Callback Verification: PayHere validates credentials and triggers the onCompleted callback.

7. Fulfillment & Stock Decrement: The backend transitions the order status to 'paid', decrements inventory stock atomically, and redirects the user to the Order Confirmation page.

8.3 Security Hash Generation Mechanism

The backend calculates the verification hash using Node.js's native crypto module according to PayHere's security specification:

Formula: hash = MD5(merchant_id + order_id + amountFormatted + currency + MD5(merchant_secret))

To prevent floating-point precision mismatch, amounts are strictly formatted to two decimal places (amount.toFixed(2)). The merchant secret is stored securely on the server and is never exposed to the client browser.

8.4 Order Status Lifecycle

The order fulfillment lifecycle is tracked through five progressive statuses:

- **Pending:** Order created in database, awaiting customer payment.
- **Paid:** Payment verified through PayHere; product stock quantities decremented.
- **Processing:** Store administrator packs accessories and generates dispatch airway bills.

• **Shipped:** Order handed over to courier service with tracking consignment number.

• **Delivered:** Package safely delivered and signed for by the customer.

9. Testing and Quality Assurance Results

9.1 Usability Testing & UX Audits

Usability testing was conducted with a representative group of five target users to assess task completion rates, interface clarity, and friction points:

• **Search & Filter Efficiency:** All users located a specific phone case and fast charger within 15 seconds using the top search bar and category chips.

• **Input Validation Clarity:** Users noted that the real-time green checkmarks (✓) gave them high confidence that their phone number and delivery address were properly entered.

• **Cart Resilience:** Users tested closing their browser with items in the cart; upon reopening, 100% of items and quantities were restored from localStorage.

9.2 Automated End-to-End Testing (Puppeteer QA)

To guarantee rigorous software quality assurance, an automated browser testing suite was implemented using Puppeteer (scripts/run_qa_verification.cjs). The script launches headless Chrome instances, emulates realistic mobile viewports, executes user workflows, and records photographic evidence to document/evidence/.

9.3 Test Case Execution Matrix

The results of the automated and manual test cases are summarized in the execution matrix below:

Test Scenario	Execution Procedure	Expected Result	Status
375px Mobile Viewport Audit	Emulate 375px width in headless Chrome; inspect overflow.	Zero horizontal scroll bar; touch targets >= 48px.	**PASSED**
Real-Time Input Validation	Enter valid and invalid phone numbers & emails into form.	Instant green border & checkmark (✓) on valid input.	**PASSED**
Cart Arithmetic & Subtotals	Increment item quantity by +1 and +2; check net totals.	Subtotal and delivery charge recalculate in real-time.	**PASSED**

Cart Session Persistence	Add items to cart, trigger hard browser reload (Cmd+R).	Cart items, quantities, and totals 100% preserved.	**PASSED**
PayHere Gateway Precision	Send cart total to backend cryptographic hash generator.	MD5 hash generated matching PayHere two-decimal spec.	**PASSED**
Admin Product CRUD	Create new product via admin modal; verify in storefront.	Product saved to MongoDB and immediately displayed in catalog.	**PASSED**
Admin Route Guard Check	Attempt to access /admin using standard customer JWT.	Request rejected with HTTP 403 Forbidden / redirected.	**PASSED**

10. Conclusion and Future Enhancements

10.1 Project Summary

The development of nr-accessories.lk successfully fulfilled all academic and practical requirements specified in the ICT2142 E-Business Systems curriculum. The platform delivers an authentic, high-speed, and secure online shopping experience tailored specifically for the Sri Lankan smartphone accessories market. By coupling a React-based Single Page Application with a Node.js/Express REST backend and MongoDB persistence, the system demonstrates high performance, elegant UI ergonomics, and rock-solid financial security via PayHere integration.

10.2 Challenges Encountered and Solutions

1. Handling Currency Precision in Payment Hashing: PayHere rejected payment verification tokens due to subtle differences between JavaScript floating-point numbers and PayHere's expected two-decimal string format. Solution: Implemented standardized .toFixed(2) currency formatting across all backend hash calculations.

2. Mobile Viewport Overflow on Small Devices: Product comparison tables and navigation menus caused horizontal clipping on 375px mobile screens. Solution: Redesigned the mobile navigation to feature a fixed bottom tab bar (MobileTabBar.tsx) and an off-canvas drawer (MobileMenu.tsx).

3. Cart Synchronization Across Page Refreshes: Refreshing the page caused React state to reset to empty. Solution: Built an automatic persistence bridge linking React Context with browser localStorage.

10.3 Future System Enhancements

• **Integration with Local Courier APIs:** Connect the backend directly with third-party logistics APIs (e.g., Pronto, Domex, Koombiyo) to automatically generate tracking waybills and return real-time courier GPS tracking.

• **Buy Now Pay Later (BNPL) Integration:** Integrate localized installment payment methods (Koko and Mintpay) to enhance affordability for high-ticket charging stations and premium audio gear.

• **AI-Driven Compatibility Recommender:** Introduce an intelligent search widget where customers select their exact smartphone model (e.g., iPhone 15 Pro Max) to receive an automatically filtered, curated bundle of compatible cases, screen protectors, and fast chargers.

• **Automated SMS & WhatsApp Alerts:** Send automated dispatch confirmations and delivery PINs to customers' mobile phones upon status changes.

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